

Policy Search for Pilot-Production Viability

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Abstract

This note studies whether bounded pilot-production policies can produce a 3500-pilot force while satisfying wingman (WG), flight-lead (FL), and instructor-pilot (IP) ready-aircrew-program (RAP) requirements. The policy search is framed as a finite-horizon, open-loop control problem over a 20-year simulation. Candidate policies are generated with Sobol sampling, heuristic seeds, and Gaussian-process surrogate screening, but feasibility claims are made only after direct evaluation with the physics-backed simulator. The current-bound closeout bundle contains 2158 successful direct-physics evaluation rows across three- and five-epoch policy classes; no feasible policy was found. The best observed current-bound schedule was a three-epoch policy with $\phi = 5.83075$, requiring simultaneous relaxation of the total-pilot window, WG RAP, and FL RAP constraints. A seed-controlled input-bound study found that one-at-a-time relaxation of retention, intake, UTE, PAA, manning, FLUG, or IPUG bounds did not produce feasibility; the best such counterfactual reduced ϕ to 5.33 but still violated both inventory timing and RAP requirements. A subsequent joint widened-bound refinement, with annual intake, retention, UTE, and PAA bounds relaxed together, improved the nearest miss to $\phi = 5.63722$ but still found no feasible schedule. These results are not a proof of infeasibility, but they give concrete nearest misses and a clear trade-space basis for the next policy or requirements decision.

1 Introduction

The practical question is not whether a simulator can be tuned to improve one metric. The question is whether a policy exists that can generate the required pilot inventory and keep the force trained enough to meet RAP requirements. For this analysis, the inventory target is 3500 total pilots, and the enabled RAP constraints require zero positive WG, FL, and IP RAP shortfall during the post-assessment window.

A few terms are used throughout the note. WG, FL, and IP denote wingman, flight-lead, and instructor-pilot categories. RAP is the ready aircrew program, so a positive RAP shortfall means the simulated force lacks enough training capacity or qualified personnel to satisfy that category's configured requirement. UTE is aircraft utilization. PAA is primary aircraft authorization. FLUG and IPUG are per-phase flight-lead and instructor-pilot upgrade quotas. The assessment window begins in 2040 in this 2026–2045 simulation; results in that window, not only final-year values, determine feasibility.

The policy levers are annual B-course intake, retention rate, utilization (UTE), primary aircraft authorization (PAA), maximum manning percentage, FLUG quota per phase, and IPUG quota per phase. A policy can be static, with one value for each lever, or dynamic, with different values applied over different parts of the 20-year horizon. The dynamic policies studied here are structured open-loop schedules: the horizon is divided into a small number of contiguous epochs, and each epoch holds the seven policy levers fixed.

The search uses a two-level evidence hierarchy. Surrogates are used to explore the policy space, rank candidates, and focus direct evaluations near promising regions. Direct physics remains the authority for all feasibility statements. Sobol sequences are used for the main design-of-experiments steps because they are deterministic low-discrepancy sequences that cover bounded high-dimensional spaces more evenly than naive random sampling at the same budget. The workflow is summarized in Fig. 1.

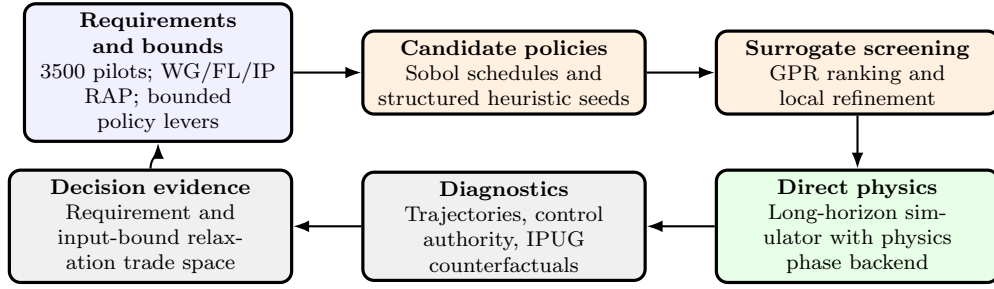


Figure 1: Viability-search workflow. Surrogates are used to propose and refine candidate policies, while direct physics is the source of truth for feasibility and for all reported nearest-miss results.

The internal phase MLP is not used as an authority in this workflow.¹ The final evaluator uses `model.phase_backend: physics`. In other words, learned models narrow the search, but every feasibility label and nearest-miss value reported below comes from the long-horizon simulator.

2 Problem Formulation

Let x_k denote the compressed force and training state at simulation phase k . The simulator state includes quantities such as total pilots, line pilots, qualification mix, staff counts, upgrade burden, and RAP shortfalls. Let u_e denote the policy controls assigned to epoch e :

$$u_e = [\text{intake}_e \quad \text{retention}_e \quad \text{UTE}_e \quad \text{PAA}_e \quad \text{manning}_e \quad \text{FLUG}_e \quad \text{IPUG}_e]^T.$$

The 20-year horizon contains 60 simulation phases. An E -epoch policy divides those 60 phases into E contiguous blocks. Within each block, the same seven controls are applied at every phase. Thus, a three-epoch policy has 21 decision variables, and each epoch is roughly one third of the 20-year horizon.

The physics-backed simulator defines the nonlinear transition map

$$x_{k+1} = f_k(x_k, u_{e(k)}),$$

where $e(k)$ maps phase k to its active epoch. The search minimizes the maximum normalized constraint violation,

$$\min_U \phi(U) = \min_U \max_i \frac{g_i(x_{0:N}; U)}{s_i},$$

where $g_i \leq 0$ is the satisfied convention and s_i is the configured constraint scale. A policy is direct-feasible only if the direct physics evaluation returns $\phi \leq 0$. The boundary $\phi = 0$ is exactly active feasibility; positive ϕ is normalized violation.

The total-pilot window constraint is

$$g_{\text{window}} = 3500 - \min_{k \in \mathcal{A}} \text{total_pilots}_k,$$

where $\mathcal{A}$ is the post-assessment window, beginning at the configured assessment year 2040. Therefore, `total_pilots_window=533` means the force falls to 2967 total pilots at the worst point in the assessment window. It does not necessarily mean the final force is short by 533 pilots.

¹The configured 16-output MLP was audited on a Sobol holdout set. Its all-target R^2 was about 0.595, and the sortie-rate-group R^2 was about 0.244. That was too inaccurate for policy claims. A later 4096-row shared ARD Matern $\nu = 2.5$ GPR improved holdout metrics, with all-target R^2 about 0.933, sortie-rate R^2 about 0.881, and simulator-rate R^2 about 0.949, but it is still used only for screening.

Table 1: Policy-input bounds used for the main direct-physics search.

Input	Type	Bounds
Annual intake	integer	10 to 350
Retention rate	continuous	0.10 to 0.65
UTE	continuous	6 to 20
PAA	integer	18 to 30
Maximum manning percentage	continuous	100 to 200
FLUG quota per phase	integer	0 to 10
IPUG quota per phase	integer	0 to 10

Table 2: Enabled requirements. Disabled requirements are not part of ϕ .

Requirement	Value
Target total pilots	3500
Allowed WG RAP shortfall	0.0
Allowed FL RAP shortfall	0.0
Allowed IP RAP shortfall	0.0
Target line pilots	disabled
Minimum experience ratio	disabled
Target staff IPs / FLs	disabled

3 Search Campaign

The main campaign used the physics-backed configuration in `outputs/viability/dynamic_policy_search/config_physics.yaml`. The three-epoch search evaluated structured heuristic seeds and a Sobol initial design, fitted a Matern GPR to direct-physics ϕ values, screened a large Sobol candidate pool with a lower-confidence-bound rule, and direct-verified the top candidates. A targeted refinement then sampled and screened local perturbations around the best direct schedules. A five-epoch search tested whether a finer open-loop schedule changed the trade space. It did not improve the best ϕ , so the planned 10-epoch expansion was not run.

Table 3: Current-bound direct-physics dynamic search summary.

Quantity	Value
Epoch classes tested	3 and 5
Simulation phases	60
Original three-epoch evaluations	615
Three-epoch refinement evaluations	256
Five-epoch evaluations	1287
Combined closeout evaluation rows	2158
Direct-feasible policies found	0
Best observed ϕ	5.83075
Best active constraint	WG RAP

4 Direct-Physics Results

No direct-feasible policy was found in the tested structured policy classes. The best observed schedule was the three-epoch refined policy `refine_0008`. It reached final inventory, but it did not reach the 3500-pilot target early enough in the assessment window and still violated WG and FL RAP.

Table 4: Best observed three-epoch policy.

Control	Epoch 1	Epoch 2	Epoch 3
Annual intake	147	144	122
Retention rate	0.5774	0.5696	0.5468
UTE	20.0	20.0	19.4119
PAA	30	30	30
Maximum manning percentage	170.324	176.677	182.110
FLUG quota per phase	2	2	3
IPUG quota per phase	0	0	0

Table 5: Positive direct violations at the best observed policy.

Constraint	Positive violation
Total-pilot window	533 pilots
WG RAP shortfall	5.83075
FL RAP shortfall	2.94844
IP RAP shortfall	none

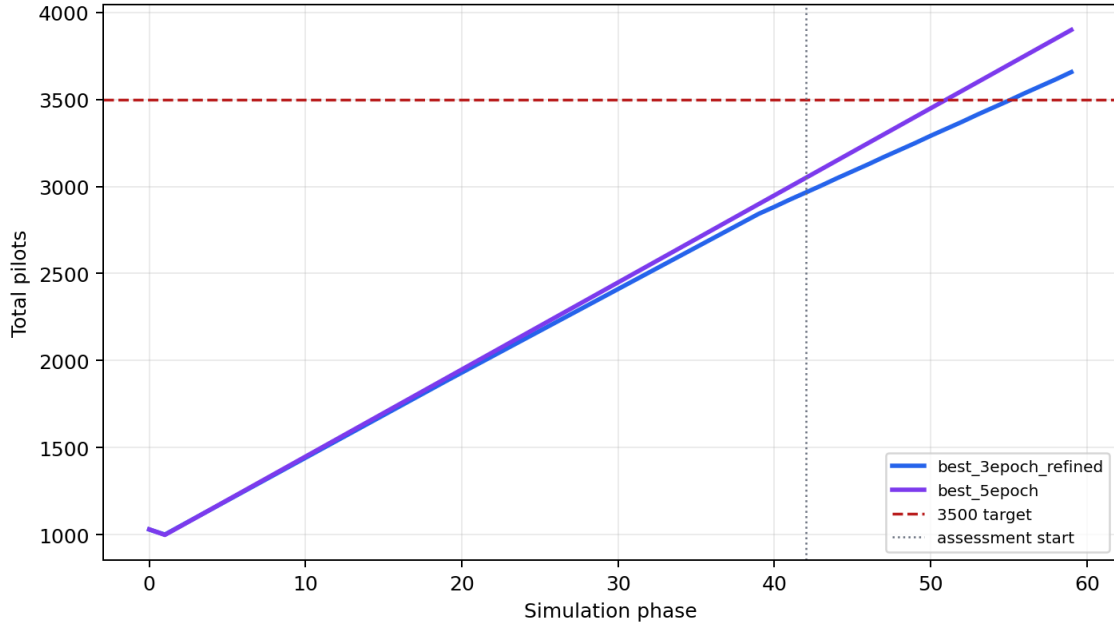


Figure 2: Total-pilot trajectories for the best three-epoch refinement and the best five-epoch schedule. Both trajectories recover final inventory, but both remain below the 3500-pilot target at the assessment start.

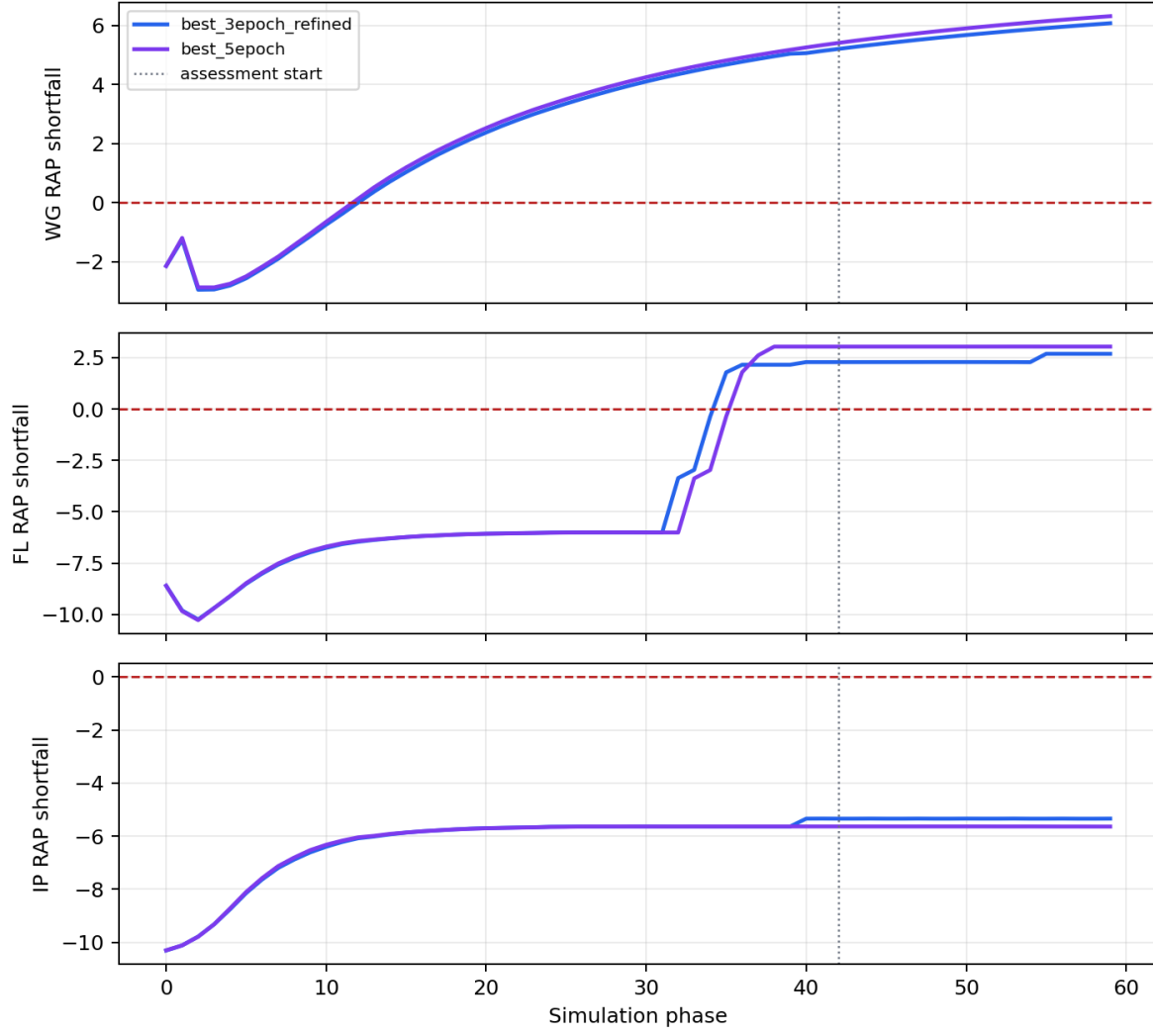


Figure 3: WG, FL, and IP RAP shortfalls for the same direct-physics schedules. Positive values violate the zero-shortfall requirement. IP RAP remains slack, while WG and FL RAP become positive during the assessment window.

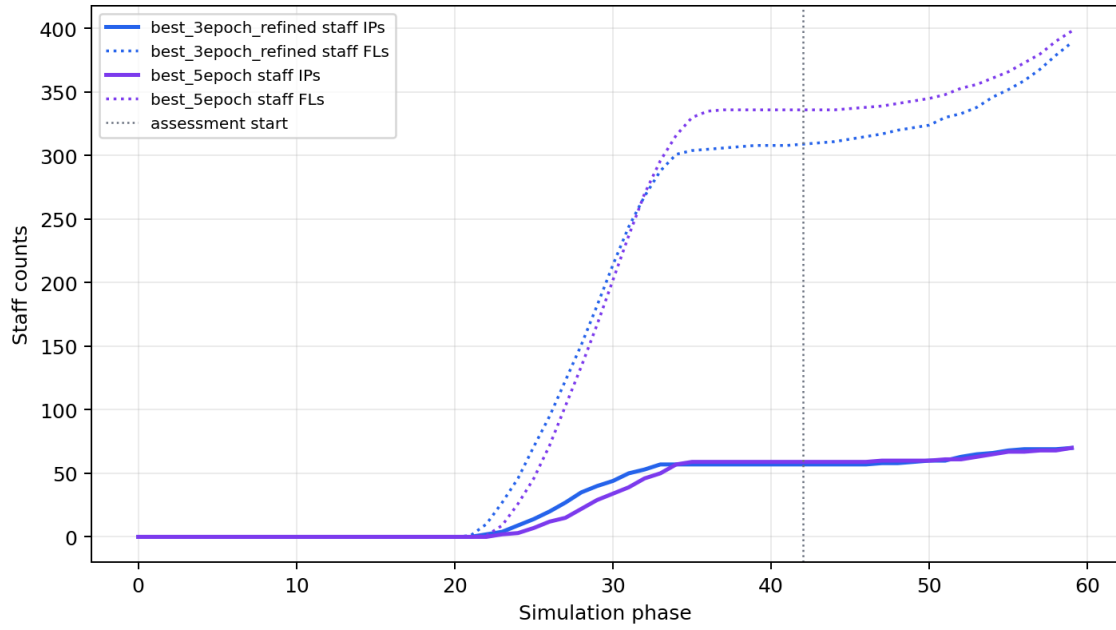


Figure 4: Staff IP and staff FL trajectories for the best three-epoch refinement and the best five-epoch schedule. The best schedules maintain staff IP inventory even with all-epoch IPUG quota set to zero, while staff FL counts increase substantially through the assessment window.

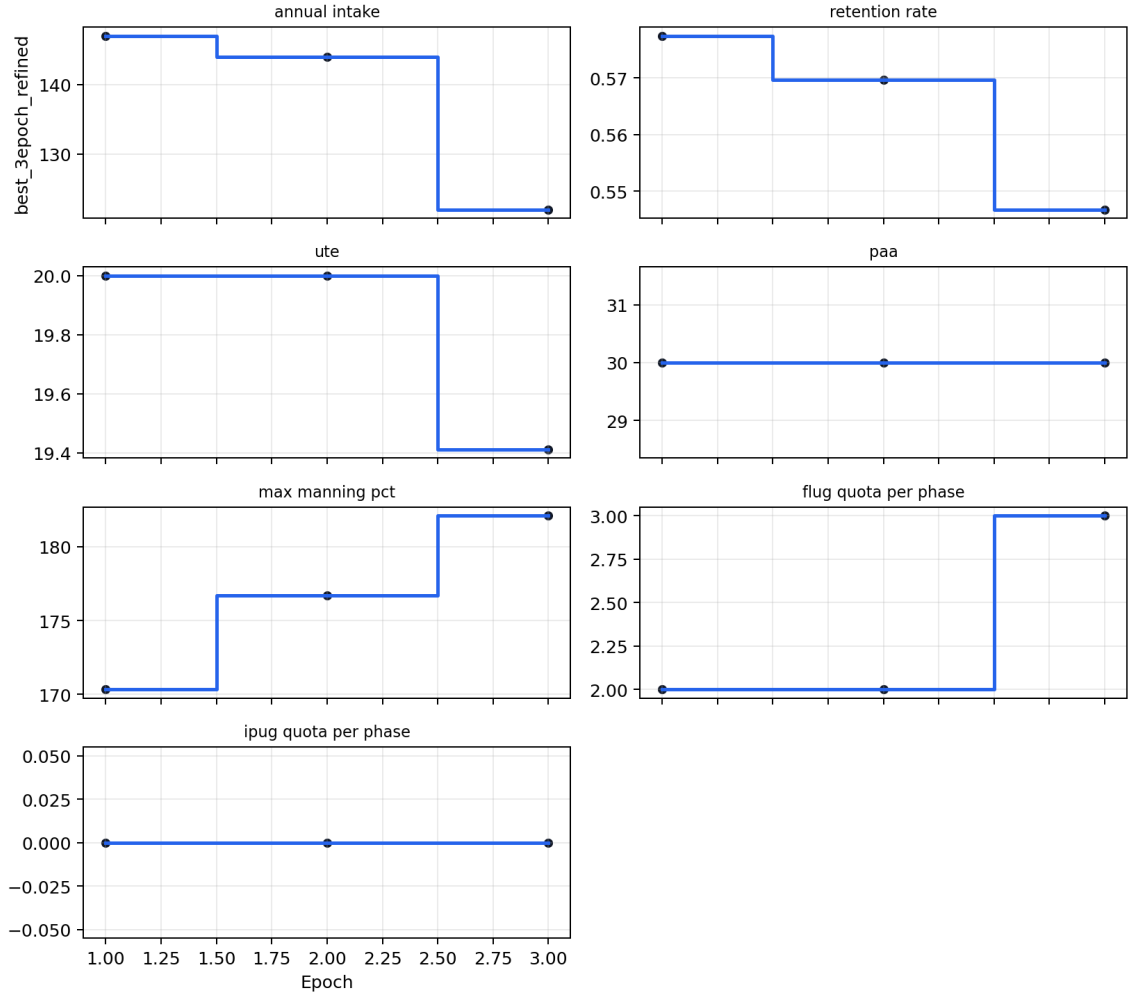


Figure 5: Epoch controls for the best observed three-epoch policy. The policy keeps UTE and PAA at or near their upper bounds, gradually reduces intake and retention, and sets IPUG quota to zero in all epochs.

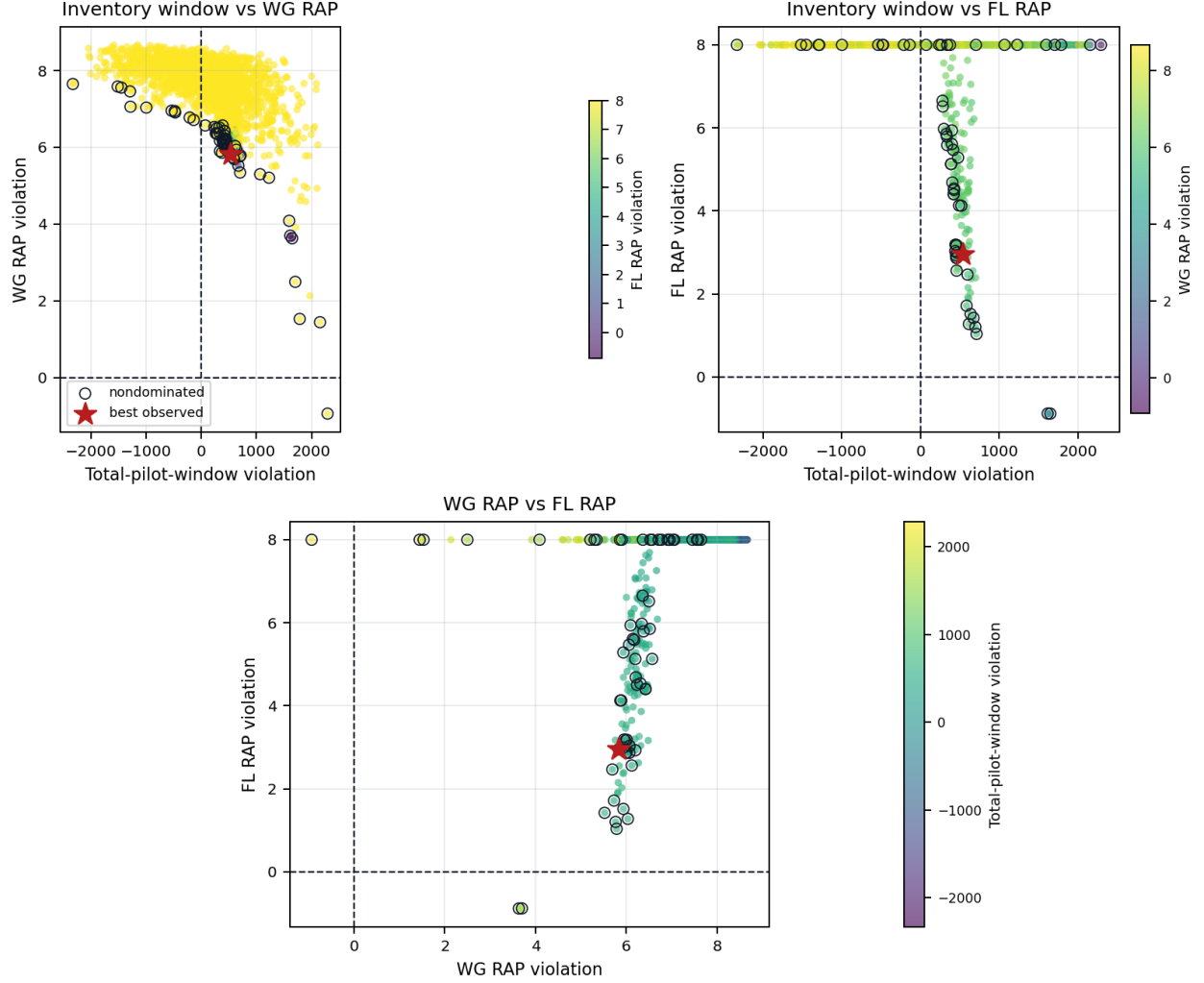


Figure 6: Three direct-physics views of the current-bound constraint trade space. Each panel shows the same evaluated policies with two violations on the axes and the third violation encoded by color. Lower-left is better; dashed lines mark zero violation. Open circles mark policies that are nondominated over total-pilot-window, WG RAP, and FL RAP violations, and the red star marks the best observed policy.

The IPUG row in Table 4 does not mean the simulated force has no instructor pilots. With IPUG quota set to zero in all epochs, the best policy still ends with 71 staff IPs, has a minimum post-assessment staff-IP count of 58, and has IP RAP slack of 5.15. Figure 4 shows the same point over the full horizon. The optimizer is effectively using existing IP inventory and pipeline dynamics while avoiding the FL RAP penalty associated with additional IP upgrades.

The active-constraint distribution reinforces this joint-trade interpretation. Across the 2158 current-bound closeout evaluation rows, the active constraint was FL RAP 1128 times, WG RAP 731 times, and total-pilot window 299 times. Each core constraint was satisfied by at least one policy somewhere in the search, but the search did not find a policy satisfying the inventory window, WG RAP, and FL RAP simultaneously.

5 Requirement and Input-Bound Relaxations

The existing requirement-relaxation study combines the direct three-epoch, three-epoch refinement, and five-epoch result tables. The minimum observed L-infinity normalized relaxation is the same best schedule, **refine_0008**. With the current input bounds, the best observed simultaneous relaxation is 533 pilots in the assessment window, 5.83075 WG RAP shortfall, and 2.94844 FL RAP shortfall. IP RAP does not need relaxation at that point.

Table 6: Requirement relaxation implied by the best observed direct policy.

Requirement relaxation	Value
Total-pilot window	533 pilots
WG RAP shortfall	5.83075
FL RAP shortfall	2.94844
IP RAP shortfall	0.0
Maximum normalized relaxation	5.83075

The single-constraint minima in the relaxation study are all zero: at least one direct-evaluated policy satisfies each individual core requirement in isolation. The useful information is therefore in the combined sets. Table 7 shows that the inventory window, WG RAP, and FL RAP conflict jointly rather than through one isolated blocker.

Table 7: Best observed normalized relaxation for selected requirement sets.

Relaxed requirement set	Best schedule	Max normalized relaxation
WG RAP + FL RAP	heuristic_0005	3.63206
Total window + FL RAP	heuristic_0000	4.40232
Total window + WG RAP	refine_0008	5.83075
Total window + WG RAP + FL RAP	refine_0008	5.83075

A separate seed-controlled input-bound study widened each policy upper bound one at a time around the best schedule and reran direct physics. This is not a global reoptimization under widened bounds. It is a first, controlled answer to which single input-bound relaxations have immediate local authority. No one-at-a-time bound relaxation produced feasibility. Raising UTE or PAA helped the RAP side of the trade but left the total-pilot-window violation binding at 533 pilots. Increasing annual intake fixed inventory timing in some cases, but it drove FL RAP to the clipping level of 8.0. Increasing retention above the current 0.65 cap did not solve the RAP constraints in this fixed-shape sweep.

The IPUG-specific counterfactual made the interpretation sharper. Holding the best schedule fixed and setting IPUG quota to 0, 2, 5, 8, 10, 15, or 20 in all epochs found the best result at IPUG=0. Any positive all-epoch IPUG value drove FL RAP to 8.0 while leaving the total-pilot window and WG RAP violations unresolved. This does not prove IPUG should never be used in a reoptimized policy, but it explains why the current best local policy sets IPUG to zero.

Table 8: Seed-controlled one-at-a-time input-bound counterfactuals. Positive total-window values are assessment-window inventory violations; negative values mean surplus relative to the 3500-pilot target.

Counterfactual	Best ϕ	Total-window g	WG RAP	FL RAP
Baseline best schedule	5.83075	533	5.83075	2.94844
PAA upper bound to 40	5.33	533	5.19227	1.03900
UTE upper bound to 25	5.33	533	5.19785	1.86937
Retention upper bound to 0.95	5.83075	517	5.83075	3.09891
Annual intake upper bound to 400	8.0	-132	6.89748	8.0
FLUG upper bound to 20	5.79383	533	5.79383	3.15091
IPUG upper bound to 20	8.0	536	6.16600	8.0

6 Local Control Authority

A finite-difference diagnostic was also run around the best schedule. The diagnostic asks a local question: if one epoch control is nudged near the best policy, which constraints move? It is not a controller design and it is not a substitute for direct verification.

Table 9: Dominant local control authority near the best trajectory.

Response	Highest-authority control	Sensitivity
ϕ	Epoch 3 IPUG quota	1.920
Total-pilot window	Epoch 1 retention rate	90.91
WG RAP	Epoch 2 retention rate	1.587
FL RAP	Epoch 2 retention rate	6.005
IP RAP	Epoch 2 retention rate	-3.679

The diagnostic is cautionary. Retention has strong local authority, but not in a uniformly helpful direction: it can help one RAP measure while worsening another or while changing the inventory window. IPUG has authority over IP inventory, but near the best schedule it worsens the active objective through FL RAP. This supports a combined trade study rather than a single-knob fix.

7 Joint Input-Bound Reoptimization Check

The one-at-a-time bound study is intentionally local, so a final targeted check jointly widened the most relevant upper bounds and reran the dynamic search machinery. The widened configuration set annual intake high to 400, retention high to 0.85, UTE high to 25, and PAA high to 40. Maximum manning, FLUG, and IPUG bounds were left at their current values because the preceding diagnostics did not identify them as promising first-order relaxations.

Two runs were made under the joint-bound configuration. The first was a fresh three-epoch dynamic search with 519 initial direct evaluations and 128 optimizer-selected direct verifications. The second was an anchored refinement from the current-bound best schedule, with 128 new direct verifications and the previous 871-row current-bound refinement bundle retained for comparison. This is still observed-search evidence, not a proof over the widened design space.

Table 10: Targeted joint-bound reoptimization check. Positive total-window values are assessment-window inventory violations. IP RAP was slack in all listed nearest misses.

Case	Rows	Feasible	Best ϕ	Total window	WG RAP	FL RAP
Current-bound refined bundle	871	0	5.83075	533	5.83075	2.94844
Joint-bound fresh search	647	0	6.00173	456	6.00173	2.86928
Joint-bound anchored refinement	999	0	5.63722	534	5.63722	4.91042

The anchored joint-bound refinement improved the maximum normalized violation from 5.83075 to 5.63722, but it did not remove the joint conflict. The best widened-bound point reduced WG RAP violation while keeping the inventory-window violation essentially unchanged and increasing FL RAP violation. In other words, relaxing these input bounds together created a better nearest miss, not a feasible policy.

8 Limitations and Recommendations

The search is direct-physics evidence, not a mathematical infeasibility proof. It covers structured three- and five-epoch open-loop schedules under the bounds in Table 1. It does not cover fully free per-phase controls, closed-loop feedback, or an exhaustive widened-bound optimization. The one-at-a-time input-bound study shows that no single tested bound change solved the problem around the current best schedule. The targeted joint-bound check in Table 10 shows that widening the most relevant input bounds together improved the nearest miss but still did not produce feasibility.

The decision implications are:

1. If policy input bounds remain fixed, the best observed requirements trade starts from 533 pilots in the assessment window, 5.83075 WG RAP shortfall, and 2.94844 FL RAP shortfall.
2. If requirements remain fixed, the completed joint-bound check suggests that modest widening of intake, retention, UTE, and PAA is insufficient by itself. Any further input-bound search should either broaden the bound set or increase the policy-class richness, rather than repeat another one-at-a-time sweep.
3. If neither requirements nor input bounds can change, the next justified search class is richer dynamic policy structure. A fully free per-phase schedule should be considered only after deciding that the added complexity is worth the interpretability cost.
4. The dashboard and report should center the direct-physics dynamic result bundle, nearest misses, and relaxation tables. Legacy static sliders should remain secondary screening tools.

9 Conclusion

Under the current tested policy bounds and structured open-loop policy classes, the direct-physics search did not find a feasible policy. The nearest miss is clear: the force eventually exceeds 3500 pilots, but not soon enough in the assessment window, and WG/FL RAP constraints remain positive. The first input-bound relaxation study did not reveal a single simple bound change that creates feasibility. The targeted joint widened-bound refinement improved the nearest miss but still did not create feasibility. The evidence therefore points to a joint trade: relax the inventory/RAP requirements, broaden the input-bound or policy-class search beyond the targeted check here, or expand the dynamic policy class with a clear reason for the added complexity.

A Reproducibility Notes

The main artifacts used in this note are:

- current-bound three-epoch refinement and five-epoch search bundles;
- requirement-relaxation, one-at-a-time bound-relaxation, and IPUG counterfactual outputs;
- trajectory and trade-space figure artifacts used for the paper figures;
- the generated joint-bound config and its fresh-search and anchored refinement result bundles.

These output directories and generated configs are analysis artifacts and are intentionally ignored by git. The tracked repository contains the curated publication figures under `docs/figures/viability/`, the source code used to regenerate the analysis, and this compiled note. The dashboard defaults to the ignored local output tree; on a clean checkout, regenerate the artifacts below or point the sidebar paths at an equivalent local result bundle.

The trained single-phase GPR used during surrogate screening is also a generated artifact. The canonical shared ARD Matern $\nu = 2.5$ bundle from this campaign is about 129 MB, so it is not tracked in normal git. If long-term preservation of that model is required, it should be stored through Git LFS or an external artifact store with a checksum and a small repository manifest. The direct physics evaluations remain the source of truth for all claims in this note.

The joint-bound config is the baseline physics config with four upper-bound changes: annual intake 350 to 400, retention 0.65 to 0.85, UTE 20 to 25, and PAA 30 to 40. Other policy bounds are unchanged.

The closeout artifacts can be regenerated with:

```

ROOT=outputs/viability/dynamic_policy_search
BASE_CFG=$ROOT/config_physics.yaml
BASE_REFINE=$ROOT/run_3epoch_refine_2048_131072_256/all_evaluations.parquet
BASE_SENS=$ROOT/run_3epoch_refine_2048_131072_256/diagnostic/local_sensitivity.csv
FIVE_EVALS=$ROOT/run_5epoch_1024_131072_256/all_evaluations.parquet
JOINT_CFG=$ROOT/config_physics_joint_bounds_paa40_ute25_ret085_intake400.yaml
JOINT_SEARCH=$ROOT/run_3epoch_joint_bounds_paa40_ute25_ret085_intake400_512_65536_128
JOINT_REFINE=$ROOT/run_3epoch_joint_bounds_refine_paa40_ute25_ret085_intake400_1024_65536_128

venv/bin/python -m src.viability.cli dynamic-relaxation-study \
  --config $BASE_CFG \
  --evaluations $BASE_REFINE \
  --evaluations $FIVE_EVALS \
  --output-dir $ROOT/relaxation_study_v1

venv/bin/python -m src.viability.cli dynamic-bound-relaxation-study \
  --config $BASE_CFG \
  --evaluations $BASE_REFINE \
  --output-dir $ROOT/bound_relaxation_v2 \
  --epochs 3 --sweep-points 5 --workers 8 --checkpoint-every 10

venv/bin/python -m src.viability.cli dynamic-ipug-diagnostic \
  --config $BASE_CFG \
  --evaluations $BASE_REFINE \
  --output-dir $ROOT/ipug_counterfactual_v2 \
  --epochs 3 --ipug-values 0,2,5,8,10,15,20 --workers 8 --checkpoint-every 10

venv/bin/python -m src.viability.cli dynamic-trajectory-artifacts \
  --config $BASE_CFG \
  --output-dir $ROOT/paper_artifacts_v1 \
  --evaluation $BASE_REFINE \
  --epoch-count 3 --label best_3epoch_refined \
  --evaluation $FIVE_EVALS \
  --epoch-count 5 --label best_5epoch

venv/bin/python -m src.viability.cli dynamic-search \
  --config $JOINT_CFG \
  --output-dir $JOINT_SEARCH \
  --epochs 3 --initial-samples 512 --optimizer-pool-size 65536 \
  --verify-top 128 --workers 8 --checkpoint-every 25

venv/bin/python -m src.viability.cli dynamic-refine \
  --config $JOINT_CFG \
  --previous-evaluations $BASE_REFINE \
  --diagnostic-sensitivity $BASE_SENS \
  --output-dir $JOINT_REFINE \
  --epochs 3 --local-samples 1024 --optimizer-pool-size 65536 \
  --verify-top 128 --workers 8 --checkpoint-every 25

```

References

- [1] *Internal Phase Backend Audit and Surrogate Plan*. Repository document: docs/internal_phase_surrogate_audit_plan.md.
- [2] *Viability Physics Evaluation Performance Plan*. Repository document: docs/viability_physics_performance_plan.md.

[3] *Dynamic Policy / Finite-Horizon Control Search*. Ignored diagnostic artifact:
outputs/viability/dynamic_policy_search/run_3epoch_refine_2048_131072_256/diagnostic/
dynamic_control_report.md.